

Coursework cover sheet**Section A - To be completed by the student**

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Penalties: No late work will be accepted. If you are unable to submit coursework on time due to extenuating circumstances, you may be eligible for an extension. Please consult the lecturer.	
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Section B - To be completed by the module leader

Intended learning outcomes assessed by this work (5023CMD): 1. Adhere to the social and legal implications of artificial intelligence. (A4, PLO11). 2. Identify the various approaches to knowledge representation in artificial intelligence. (C4, PLO1) 3. Assess problem-solving strategies under uncertain conditions, maintaining control and accuracy in execution. (P5, PLO3) 4. Illustrate an understanding of knowledge-based systems, and their respective architectures. (C4, PLO1) 5. Demonstrate artificial intelligence approaches in practical tasks to determine the most effective solution for an intelligent application. (P5, PLO3) 6. Verify human factors and the impact on Artificial Intelligence. (A5, PLO10)		
Marking scheme	Max	Mark
1. Assignment 1	100	
Total	100	
Lecturer's Feedback		
Internal Moderator's Feedback		

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List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
HR	Human Resource
OECD	Organisation for Economic Co-operation and Development
PETs	Privacy-Enhancing Technologies
BIPA	Biometric Information Privacy Act
LLMs	Large Language Models
CV	Curriculum Vitae
PDPA	Personal Data Protection Act
MCMC	Malaysian Communications and Multimedia Commission
EU	European Union
GDPR	General Data Protection Regulation
SQL	Structured Query Language
ESCO	European Skills, Competences, Qualifications and Occupations

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List of Mathematical Symbols

Naïve Bayes Classifier

Symbol	Definition	Context of Application
$P(A B)$	Conditional probability	Evaluates the likelihood of candidate success given extracted skill sets.
$\propto$	Proportional to	Establishes a relative ranking metric while omitting the constant denominator $P(Success)$.
Π	Product Operator	Computes the joint likelihood across all independent candidate skills.
i	Skill Indexer	Represents a single and different skill attribute parsed from the candidate's profile, where $i \in \{1, 2, \dots, n\}$.

Successive Beam Expansion and top-k Selection Logic

Symbol	Definition	Context of Application
B_t	Active Beam Set	Represents the prioritized subset of candidate records retained at the current stage t .
top- k	Selection Function	A filtering operation that selects exactly k elements with the highest heuristic scores from a set.
y_{t-1}	Partial Sequence	A sequence of attributes or candidates selected in the preceding stage.
$\cup$	Set Union Operator	Represents the expansion step where a new candidate record or skill attribute is appended to an existing path.
$\{c\}$	Candidate Element	An individual candidate profile is being evaluated for insertion into the beam.
$\in$	Set Membership Operator	Indicate the element that a specific path or candidate belongs to a designated set.
B_{t-1}	Proceeding Beam Set	The candidate survival pool was carried over from the previous iteration.
$\mathcal{C}$	Total Candidate Space	The finite pool of eligible candidate profiles that passed initial rule-based screening.

Log-Probability Heuristic Scoring for Candidate Ranking

Symbol	Definition	Context of Application
Score(c)	Heuristic Fitness Score	The quantitative value assigned to a candidate c to determine ranking priority in the beam.
$\sum_{i=1}^n$	Summation Operator	Instructs the system to add up a sequence of values starting from index i up to the final feature n .
n	Total Profile Picture Count	The total number of skills that identified in candidate c .
S_i	Individual Skill Variable	The specific $i - th$ skill attribute extracted from the candidate profile being cross-referenced.

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1.0 AI in the Recruitment Domain

Chosen Scenario: AI in Recruitment and Automated Hiring

Real-World Applications: Workday automated screening, Unilever/Pymetrics cognitive assessments, HireVue algorithmic auditing and Amazon automated bias screening.

AI usage is in high demand across modern industries. Approximately 88% of companies use AI tools during recruitment process to select candidates from large applicant pools (Amitabh, U. ; Ansari, A, 2025). These algorithms are trained on historical data that represents past prejudice rather than future potential which puts underlying discrimination at scale although offer efficiency. In terms of social, legal and ethical aspects, this essay suggests that AI hiring tools generate a basic tension between the benefit of the applicant and the efficiency of the company.

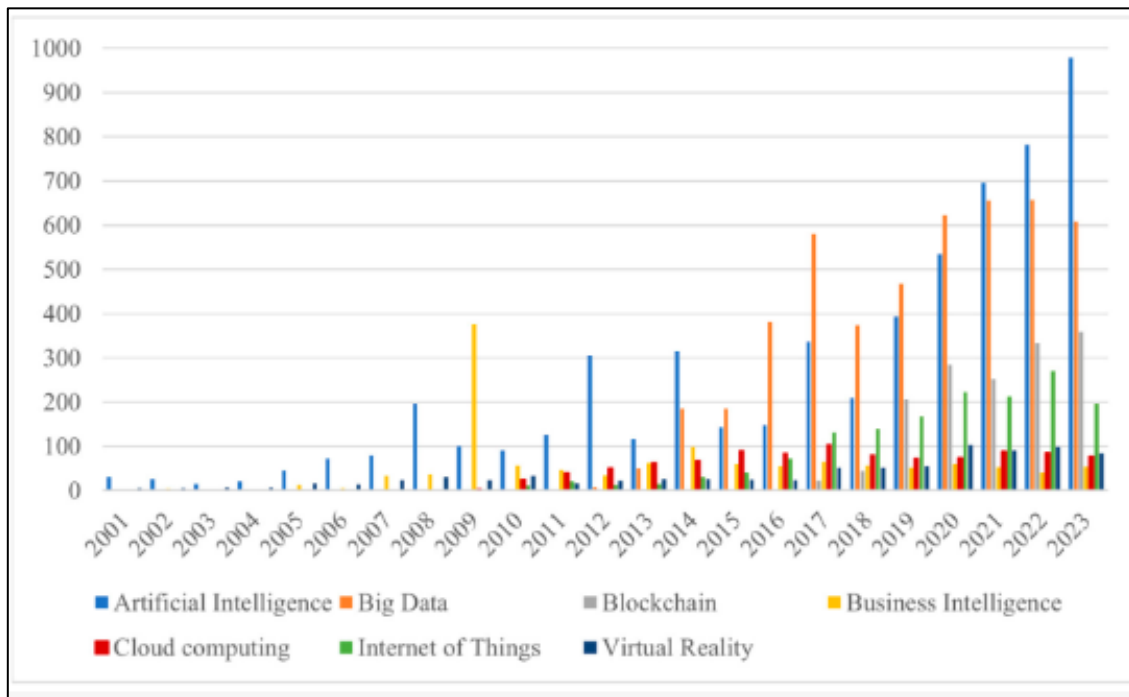


Figure 1.1 Comparison of the number of publications defined by modern ICT for the years 2001–2023 (Koman, Gabriel; Boršoř, Peter; Kubina, Milan, 2024)

1.1 Social Implications

1.1.1 Recruitment Efficiency and Socioeconomic Bias

Companies increasingly adopt AI-driven predictive modelling to improve recruitment efficiency by identifying high-performing candidates through proxies such as university background and geographical location which can help recruiters to process large volumes of applications more rapidly. These systems can handle thousands of applications in a much shorter time than manual screening. Based on a 2024 meta-analysis of 87 empirical studies, AI implementations in HR deliver an average productivity boost of 23% and cost savings of 31% compared to manual approaches (Henderson, J. ; Liu, W., 2024). From a business perspective, AI recruitment tools reduce time-to-hire, cut down administrative workload and improve scalability in large-scale hiring processes.

However, such algorithms may reproduce existing social disadvantages by prioritising historical patterns of privilege over individual potential, even among applicants with equivalent skills. Consequently, access to elite educational institutions may be incorrectly interpreted as an indicator of competence which reflecting socioeconomic privilege rather than merit. The 2022- 2023 Workday lawsuit shows AI in hiring may cause unfair outcomes because automated screening tools can reject qualified candidates based on factors like race, age and disability (Shumway, 2024). This raise concerns that qualified candidates may be rejected without being given the opportunity to explain employment or educational gaps caused by socioeconomic factors. As a result, rejected applicants may experience discrimination without having sufficient transparency to meaningfully challenge automated decisions effectively.

Although predictive modelling improves recruitment efficiency, reliance on socioeconomic proxies risks institutionalising structural inequality under the appearance of technological objectivity.

1.1.2 Bias Reduction and Bias Replication in AI Hiring

Automated hiring systems are often promoted as mechanisms for reducing subjective human bias through standardized candidate evaluation. AI recruitment system can be designed to prioritise measurable indicators of skills, cognitive ability and behavioural traits while minimising direct exposure to demographic identifiers such as gender, ethnicity or physical appearance. For example, Unilever has adopted this approach by partnering with Pymetrics to implement cognitive games assessments supported by machine learning to evaluate candidates based on behavioral analysis and cognitive data rather than relying traditional hiring methods (Marr, n.d.). It has resulted in a 16% increase in the ethnic diversity in new hires (Unilever, n.d.). Therefore, AI-supported screening systems can reduce this form of discrimination in the early screening stage.

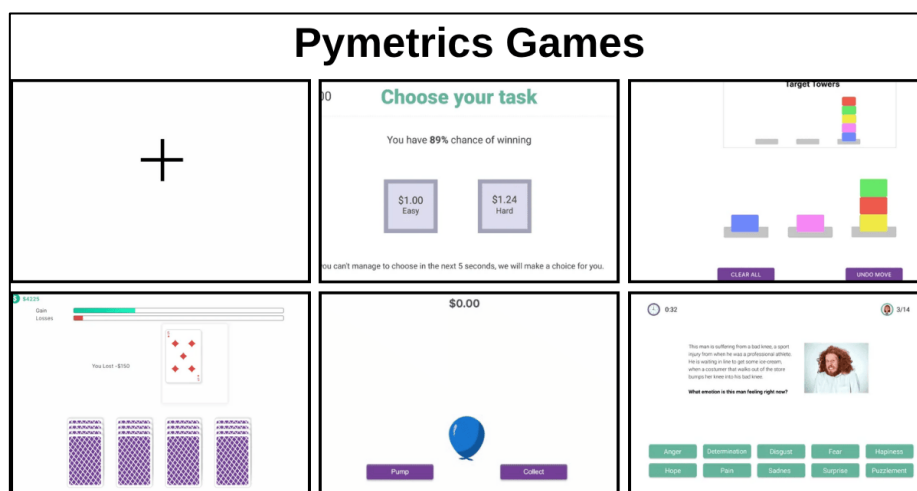


Figure 1.2 Examples of Gamified Cognitive Assessments (JobTestPrep, n.d.)

In contrast, some recruitment algorithms learn patterns from past hiring decisions. If employment practices in the past were biased toward a particular group of people, the AI will replicate the patterns and favored particular demographic group at scale. Consequently, minority applicants are disadvantaged when algorithms penalize the variation that does not match the system's standard. For example, Amazon's 2018 scandal, where its AI screening tool was found to be biased against women as it had been trained on a decade of hiring data that predominantly reflected male hires (Dastin, 2022). Critically, the system did not intentionally choose to discriminate as it identified historical male hiring patterns as indicators of successful candidates and reproduced them algorithmically. Therefore, although AI-driven assessments may reduce individual recruiter subjectivity, AI can simultaneously amplify historical discrimination systematically across large applicant populations.

This demonstrates that algorithmic objectivity does not necessarily eliminate discrimination as biased training data may cause AI systems to systematically reproduce institutional prejudice at scale.

1.2 Legal Implications

1.2.1 Regulatory Compliance and Accountability Auditing

Standardized AI recruitment tools can strengthen organizational compliance by providing structured and auditable hiring processes that are more predictable than manual hiring decisions. On the positive side, standardized AI systems can document the process of screening, ranking and evaluation of applicants, which can help employers to demonstrate adherence with anti-discrimination standards more systematically. For instance, HireVue made a public commitment to algorithmic audits and collaborated with a third-party organization, O'Neil Risk Consulting to its tools for adverse impact and fairness (HireVue, 2021). Similarly, New York City's Local Law 144 legally requires companies to conduct independent bias audits of AI hiring tools and publicly disclose results (Zev Eigen J.D.; Ph.D.; Brenna D. Kelly; Danny Tobey; Ashley Carr; Kyle Kloeppel ;Jack Mansur , 2026).

However, compliance audits do not necessarily guarantee a sign of substantive fairness. In reality, many organizations continue to struggle with accountability as enforcement mechanisms remain limited and proprietary algorithms are often protected as commercial intellectual property. Consequently, rejected job applicants are often unable to determine whether automated discrimination occurred or how hiring decisions were made. This creates a huge accountability gap where legal responsibility becomes diffused between employers, AI vendors and third-party developers.

Therefore, while standardized auditing mechanisms may improve regulatory compliance and procedural consistency, limited transparency and fragmented accountability continue to challenge the legal governance of AI-driven recruitment systems.

1.2.2 Data Privacy, Surveillance and Biometric Protection

AI recruitment tools offer both valuable data governance benefits and significant privacy risks. Compared to traditional manual hiring practices, well-designed AI tools able to encrypt candidate data more consistently which reduces risks associated with fragmented record management and human error. The OECD has highlighted that modern AI systems can incorporate PETs such as encrypted data processing and federated learning to ensure sensitive information remains protected even during complex analysis (OECD, 2024). In practice, such technologies can strengthen organizational data security while improving efficiency in recruitment administration.

Conversely, the large-scale digitalization of recruitment also introduces risks of surveillance, data misuse and unauthorized profiling. AI hiring platforms frequently collect extensive behavioral and biometric data, such as facial expressions, speech patterns and personality indicators, often from third-party software suppliers. This creates vulnerabilities involving cybersecurity breaches, adversarial attacks and non-transparency data-sharing practices. HireVue, which is an AI video interviewing platform, had been under Illinois' BIPA scrutiny for its collection of candidates' facial geometry data during recorded interviews without sufficiently transparent disclosure of how that data was stored, used or shared (Shaak, 2022). This case directly illustrates the tension of how technologies marketed as efficient and objective may simultaneously collect personal biological data without applicant's informed consent.

1.3 Ethical Implications

1.3.1 Transparency

Compared to traditional hiring practices, AI recruitment systems may enhance procedural transparency by enabling hiring decisions to be systematically documented, monitored and audited at scale. AI systems generate standardized evaluation processes that allow organisations to assess hiring outcomes more consistently whereas human recruiters may apply subjective and inconsistent reasoning. This auditability may strengthen accountability by allowing discrimination patterns to be identified across a large scale of applicants. One such example is algorithmic auditing which is implemented by HireVue, which using structured evaluation systems to standardise candidate assessment and improve consistency in early-stage screening (HireVue, 2021).

But in the ethical sense, transparency in AI hiring goes beyond just auditing. However, ethical transparency extends beyond technical auditability as applicants may still receive no understandable justification for rejection decisions, thereby impairing applicants' ability to challenge outcomes and undermining principles of procedural justice. Furthermore, limited understanding of how algorithmic decisions are produced may displace human moral responsibility as untransparent systems make accountability increasingly difficult to assign. For example, research by (Chen et al., 2026) stated that even when resumes are anonymised, LLMs can still infer demographic attributes from indirect cultural and linguistic markers, thereby leading to biased hiring outcomes (Chen, B.; Tan, Z.; Khoo, S.; Doan, B. N.; Liu, Z.; Chen, N. F.; Lee, R. K. W., 2026). Therefore, although AI-driven recruitment systems may improve consistency and auditability, limited explainability undermines procedural fairness by preventing applicants from meaningfully understanding or challenging automated decisions.

1.3.2 Automation Bias and Human Oversight

AI can apply standardized evaluation criteria across all applicants which potentially minimizes inconsistencies caused by individual recruiter bias. Human oversight is frequently regarded as a crucial ethical safeguard as it allows recruiters to contextualise applicant experiences, evaluate diversity considerations and identify potentially biased outcomes that automated systems may overlook. Research shows that AI-based screening tools have been proven to increase efficiency and consistency in recruitment processes, particularly when handling large-scale CV filtering where manual evaluation would be significantly slower and less standardised (Dattner, G. W.; Rossia, A. S.; Silla, S.; Silla, R. J., 2022).

However, the presence of human oversight does not necessarily eliminate automated bias. A 2025 University of Washington study found that human recruiters exhibited almost 90% of the same hiring biases as AI (Milne, 2025). This occurs because human recruiter may treat algorithmic recommendations as authoritative rather than advisory. Consequently, automation bias may reduce critical human scrutiny and allow discriminatory outcomes to persist without meaningful intervention. The ethical concern is that candidates might be denied opportunities as a result of the AI's unaddressed biases which could be unfair and violate the principles of procedural fairness. Oversight becomes confirmatory rather than corrective when recruiters rely uncritically on AI-generated shortlists.

Therefore, while human oversight is intended to function as an ethical safeguard, excessive reliance on algorithmic recommendations may transform oversight into passive confirmation rather than active accountability.

1.4 Application of AI in Malaysia and Policy Measures Taken

The adoption of AI in recruitment in Malaysia presents unique challenges due to the country's multi-ethnic population and language variety. Ethnic inequalities are likely to be produced by AI systems when these algorithms are trained using biased historical data, which may prioritise the ability to speak languages such as Mandarin as a proxy for competence. Research has revealed that non-Mandarin speakers and non-Chinese ethnic candidates have been unfairly disqualified from accessing specific role justifications before applicants have a chance to be evaluated (Abdul Gani & Ismail, 2025).

While the PDPA 2010 provide the protection of personal data, the legislation contains limited provisions relating to algorithmic discrimination and automated decision making or AI accountability. Consequently, applicants who experience discriminatory hiring practices possess limited legal right against the automated systems and obtain explanations regarding recruitment decisions. Significantly, Malaysia lacks specialised AI governance institutions with technical expertise to oversee algorithms independently.

One realistic safeguard would be Mandatory Algorithmic Impact Assessments measure. This would require vendors in Malaysia provide bias audit documentation as a condition of market entry and this must be overseen by an existing body such as the MCMC. This measure taken minimizes algorithmic bias and systemic discrimination by preventing algorithmic systems from violate the constitutional right of equality. Furthermore, the EU AI Act can serve as a reference for Malaysia to establish binding guidelines for companies that deploying AI systems (Mendis, 2024).

Ultimately, the Malaysian challenge is not only about the use of AI in recruitment but whether governance systems are capable of keeping up with technological innovation while maintaining substantive equality and accountability.

(Task 1 Word Count: 1,677 words)

2.0 Conceptual AI System

The proposed AI hiring system combines rule-based, ontological and probabilistic knowledge. Rule-based knowledge provides transparency and auditability; ontology provides semantic reasoning and proxy detection while probabilistic reasoning handles uncertainty. This hybrid architecture addresses historical bias, proxy discrimination, lack of transparency, automation bias and privacy risks identified in Task 1.

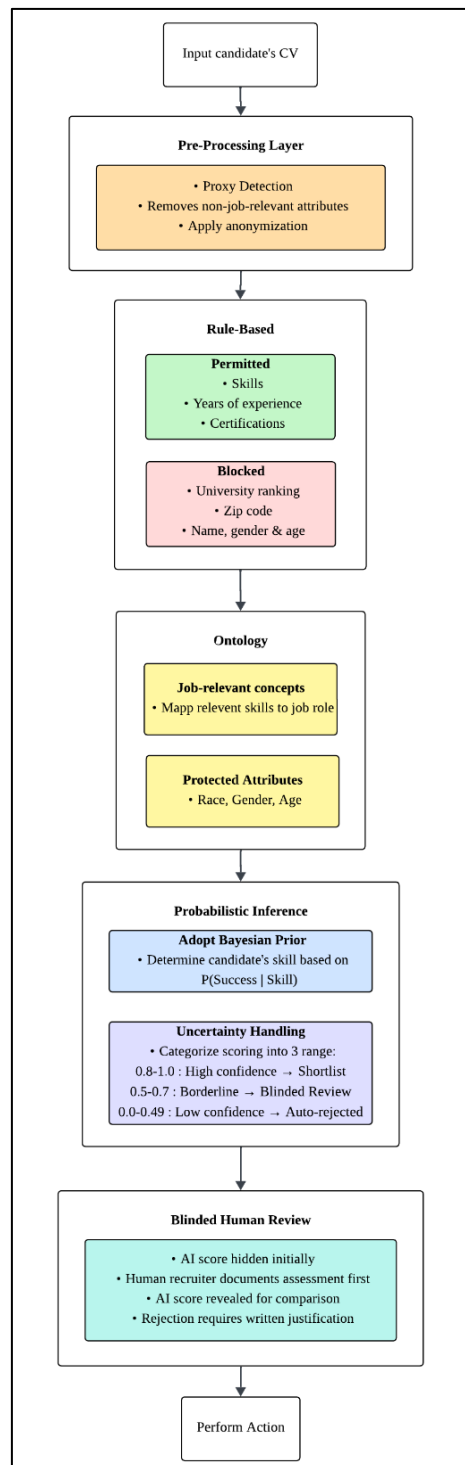


Figure 2.1 Proposed Layered AI Recruitment System

2.1 Knowledge Representation

Proxy Detection

Proxy detection is used to detect and remove indirect proxies for protected attributes during the data pre-processing stage before knowledge representation occurs. This ensures that sensitive information does not enter the rule-based, ontological or probabilistic models, thus adopting privacy by design under GDPR Article 25. All the neutralisations are logged for audit without introducing any sensitive data. A drawback is that proxy detection doesn't ensure complete neutrality as new or complex proxy relationships may evade initial detection.

Rule-Based Knowledge

The system uses symbolic rule-based knowledge representation which is suitable as the first filtering layer as recruitment compliance requires explainability and traceability (Nowak, M.; Pawłowska-Nowak, M, 2026). The encoded IF-THEN logic is applied for deterministic filtering as shown in Figure 2.2 which can be directly inspected by auditors unlike machine-learning models that are black-box and non-auditable. Permitted inputs are limited to job-relevant criteria such as skills and certifications while proxy attributes are excluded. This method limits the feature space reduces representational bias and improves legal defensibility of automated decisions.

```
IF candidate.has_required_certification = TRUE
  AND candidate.years_experience >= role.minimum_years
  AND candidate.skill_set INTERSECTS role.required_skills (min 60%)
THEN pass to ontology layer
IF candidate.attribute IN [postcode, school_name, language_dialect]
THEN neutralise (remove attribute before passing to next layer)
  AND log removal to audit trail
```

Figure 2.2 Pseudocode for Deterministic Rule-Based Filtering and Attribute Neutralization

Ontological Knowledge

The system is based on an ontology-based knowledge graph to establish the relationships between job roles, skills and competencies. As shown in the Figure 2.3, ontology able to identify SQL, data visualization and statistical analysis related to a Data Analyst role despite differing terminology. This reduces keyword dependency, improves semantic reasoning, handles vocabulary variation and increases the transferability across industries. However, ontology requires periodic updates to keep labour taxonomy aligned with the ESCO standardized frameworks (De Santo, Alessia; Malandri, Lorenzo; Mercorio, Fabio; Mezzanzanica, Mario; Nobani, Navid, 2026).

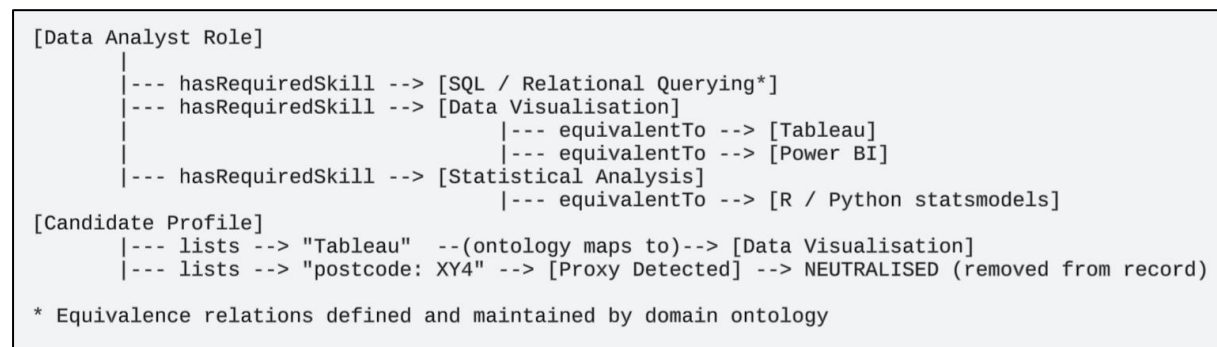


Figure 2.3 Ontological Logic Flow for Matching Candidate Attributes to Data Analyst

Probabilistic Knowledge

$$P(\text{Success} \mid \text{Skills}) \propto P(\text{Success}) \times \prod_i P(\text{Skill}_i \mid \text{Success})$$

Equation 2.1 Naïve Bayes Classifier

For candidates with resume gaps or non-linear career paths, a Naïve Bayes classifier is applied within the probabilistic knowledge model to evaluate partial evidence and reason under uncertainty. In recruitment context, Naïve Bayes classifier is suitable as it is mathematically efficient even when the number of applicants is large and produces interpretable confidence scores. It estimates job success probability using objective skill-outcome data without Amazon-style bias by excluding historical hiring decisions (Zhang, Shuo; Kuhn, Peter J., 2024). The classifier outputs a score from 0 to 1 and categorized into three decision tiers avoiding automatic rejection in case of uncertainty.

Blinded Human Review

The system flags borderline resumes for human judgment as proxy attributes have already been neutralised and removed at the ontological layer. Blinded human review mitigates automation bias by ensuring recruiters form an independent assessment before viewing any AI recommendations. This design establishes human decision authority over automated decisions as it ensures accountability is explicitly assigned rather than assumed (Zhang, 2023). However, if recruiters consistently defer to AI suggestions without independent evaluation, decision ownership becomes diffuse and the human oversight mechanism is rendered ineffective.

Ultimately, the hybrid architecture combines deterministic compliance filtering, semantic reasoning and probabilistic inference to balance efficiency with ethical safeguards. The proposed design maintains explainability, uncertainty awareness and human accountability throughout the decision pipeline, unlike purely automated black-box recruitment systems.

2.2 Reasoning or Planning Approach

2.2.1 Decision Making Under Uncertainty

Recruitment involves incomplete evidence for candidates with resume gaps or non-traditional qualifications. The system handles uncertainty through Bayesian reasoning instead of binary pass/fail logic by evaluating suitability as a probability. The probabilistic layer is used in conjunction with the deterministic rule filtering by allowing incomplete but potentially qualified candidates to remain under consideration.

Bayesian Reasoning Flow

When a candidate's record passes the rule-based filter and all proxy attributes have been neutralised during preprocessing, it enters the probabilistic reasoning stage. This computes a posterior probability of job success given the observed skills using the Naïve Bayes classifier shown in Equation 2.1.

Bayesian reasoning is highly suitable as recruitment decisions involve incomplete and uncertain information. Candidates may possess resume gaps, transferable skills or non-linear career paths that cannot be evaluated effectively using deterministic pass-fail logic. The model estimates suitability probabilistically using existing skills. This enables the system to remain exceptionally robust when processing imperfect candidate records while maintaining explainable and auditable decision-making. The final posterior score is then classified into three different tiers in Table 2.1.

Three-tier Scoring

Confidence Score Level	Interpretation	Action Taken
0.8 – 1.0	High Confidence	Auto-shortlist
0.4 – 0.79	Borderline	Escalate to blinded human review
0.0 – 0.39	Low Confidence	Auto-rejected with written skill-level justification provided

Table 2.1 Three-Tier Decision Framework for Confidence-Based Candidate Processing

The use of conditional independence of skills used in real recruitment relationships might be an oversimplification but the Naïve Bayes model is still applicable due to its interpretability and computational efficiency.

This approach is able to mitigate the risk of false-negative rejections for candidates with non-linear career paths. Crucially, a candidate who has a gap in the resume shouldn't be auto-penalised. The Naïve Bayes classifier scores only what is available. Absence of skills reduces confidence but no outright rejection is issued. Rejection will apply if the confidence score falls within low-confidence level. Borderline candidates are escalated for recruiter evaluation instead of treating uncertainty as failure.

2.2.2 Role of Human Oversight

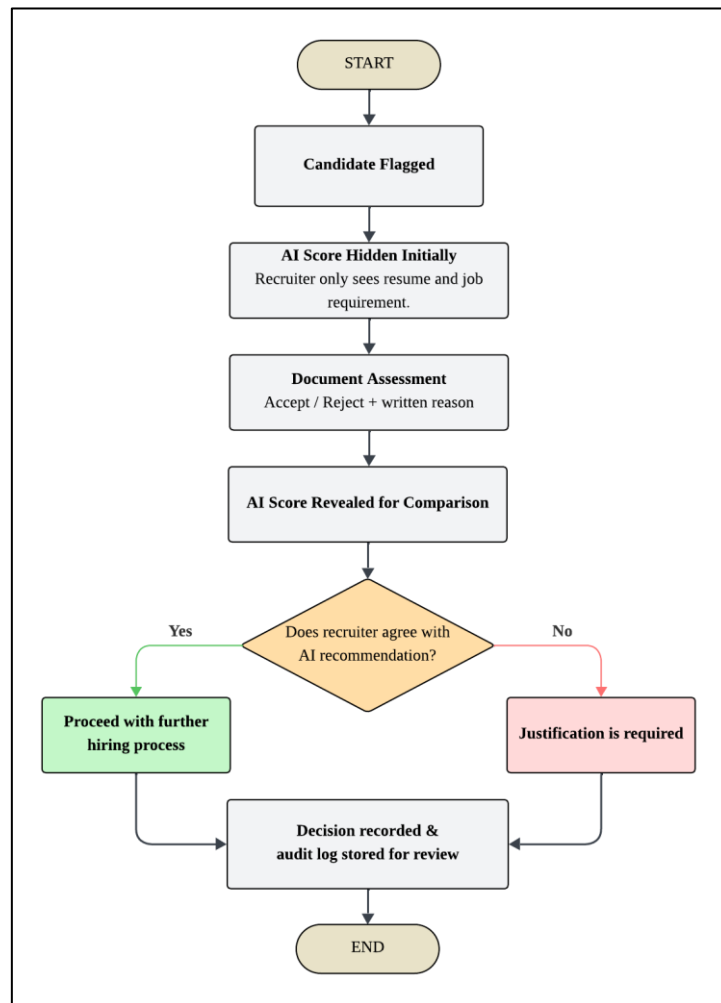


Figure 2.4 Blinded Human Review Workflow

The diagram illustrates a human-in-the-loop architecture where borderline cases are escalated for blinded review. Crucially, the AI confidence score remains hidden during the initial assessment to prevent confirmation bias where recruiter might unconsciously agree with the algorithm. Recruiters are required to form an independent assessment before the AI score is revealed. This design directly counters automation bias as recruiters cannot defer to the AI without first forming recruiters own independent evaluation.

Mandatory written reasons at the initial and rejection stages satisfy EU AI Act Article 14 as each decision is documented through audit and prevents recruiters from entirely relying on the AI score (European Parliament and Council of the European Union, 2024). Furthermore, proxy attributes have been neutralised initially and the removal of demographic indicators reduces the risk of direct discrimination during review. Even with human review, subjective bias still can be an underlying factor, therefore periodic audit and recruiter calibration are required.

2.3 Search or Inference Technique

Naïve Bayes inference is used to estimate candidate suitability under uncertainty, while Beam Search is applied to efficiently prioritise applicant processing. This combination ensures both probabilistic scoring with controlled computational complexity and scalable recruitment decision-making.

2.3.1 Naïve Bayes Inference Technique

Naïve Bayes inference is selected to handle incomplete and uncertain candidate information, especially where resumes contain missing skills, employment gaps or non-linear career paths. Naïve Bayes addresses this problem by computing posterior probabilities by treating absent attributes as non-informative for a given prediction.

The system estimates the candidate's confidence score using Equation 2.1 Naïve Bayes classifier. It allows system to compute a confidence score based on available skills instead of historical hiring patterns. For an example, Software Engineering role requires Python, SQL, C++ and JavaScript, a candidate with four skills would receive higher score and be auto-shortlisted. A candidate possessing only Python and SQL would receive a lower score and trigger human review. This avoids overly rigid rule-based filtering and supports more nuanced decision-making.

Compared to alternative techniques, logistic regression requires complete feature vectors through forcing bias-inducing imputation. Neural networks provide high predictive power but lack interpretability, limiting compliance with explainability requirements such as GDPR Article 22. Decision trees are interpretable but may overfit on small or imbalanced recruitment datasets. Naïve Bayes provides a balance between interpretability, robustness to missing data, and computational efficiency.

A minor trade-off is the conditional independence assumption where skills are treated independently despite real-world correlation. This is mitigated through careful feature engineering and probability calibration using validation data.

2.3.2 Beam Search Technique

Beam Search is used to efficiently prioritise candidates within large applicant pools under bounded computational and human review capacity. The initial filtered applicant pool forms the root level and candidates passing rule-based screening are expanded into the search space. The successful expansion and pruning logic is governed by Equation 2.2.

$$B_t = \text{top} - k (\{ y_{t-1} \cup \{c\} \mid y_{t-1} \in B_{t-1}, c \in C \})$$

Equation 1.2 Successive Beam Expansion and top-k Selection Logic

At each stage, Beam Search retains the top-k candidates based on a heuristic score derived from Naïve Bayes confidence values as formalized in Equation 2.3.

$$\text{Score}(c) = \sum_{i=1}^n \log P(s_i \mid \text{success})$$

Equation 2.3 Log-Probability Heuristic Scoring for Candidate Ranking

If a candidate receives a 0.92 confidence score, the candidate will receive high priority compared to candidate with 0.61 confidence score. The beam width k is configured based on recruiter workload capacity to balance exploration and efficiency. This makes the system adaptable to organisational resource limits.

Beam search naturally accommodates human-in-the-loop operations. Borderline candidates remain in the beam during review. If a second opinion is requested, the candidate stays in the beam while higher-confidence candidates are processed first without blocking higher-confidence candidates from progressing. This ensures that review latency does not negatively impact system throughput.

Compared to exhaustive search methods, Beam Search significantly reduces computational cost by eliminating low-confidence candidates early while still preserving the most promising subset for deeper evaluation. This alignment between algorithm design and problem structure makes beam search a stable, efficient and defensible choice.

2.3.3 Interaction Between Search and Inference Techniques

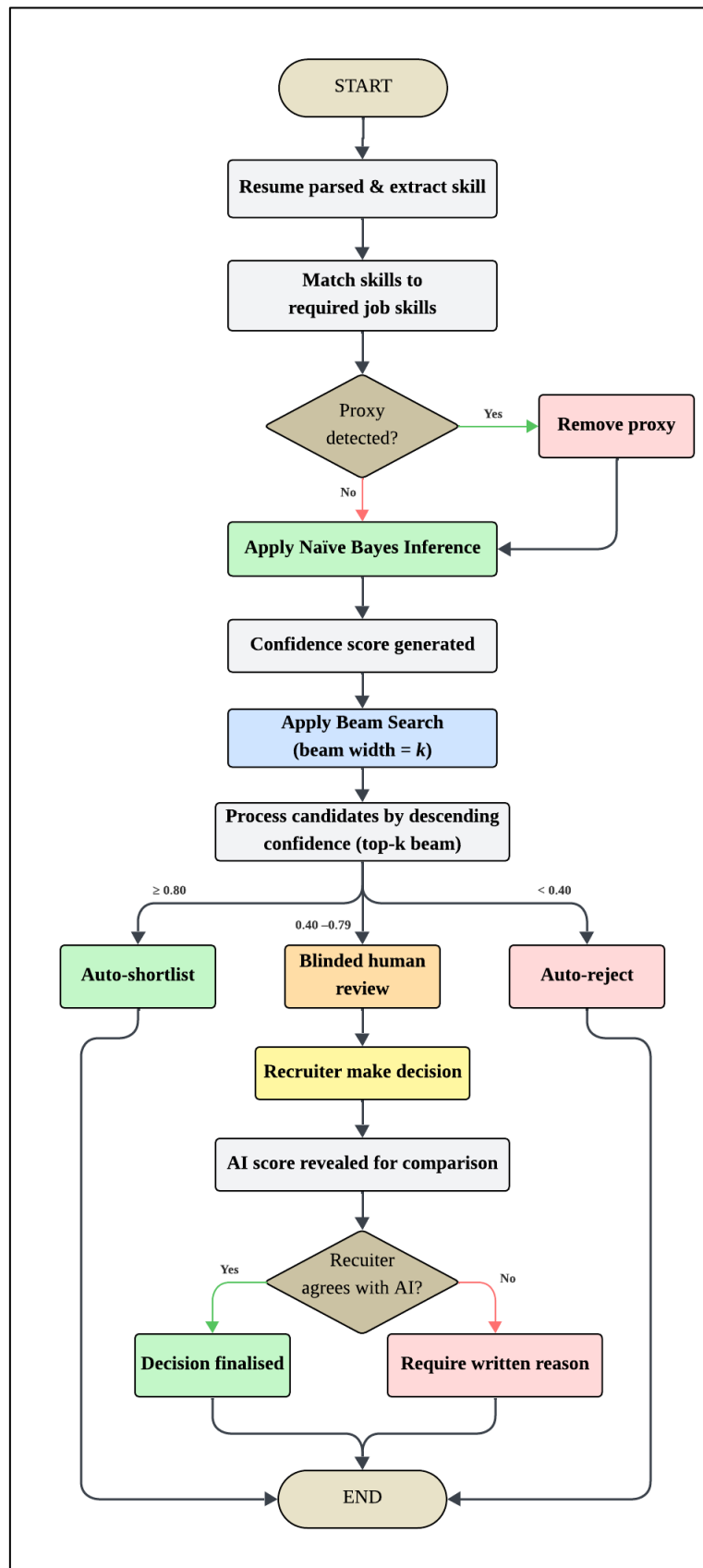


Figure 2.5 Flowchart of Candidate Prioritisation using Naïve Bayes Inference and Beam Search Heuristics

Naïve Bayes and Beam Search operate as an integrated pipeline for probabilistic ranking and efficient selection. After rule-based filtering and proxy neutralisation, Naïve Bayes computes a confidence score for each candidate. Beam Search ranks candidates by descending confidence and retains top-k in the beam by eliminating the rest.

Candidates scoring larger than 0.8 are auto-shortlisted. Candidates scoring between 0.4 and 0.79 enter blinded human review while remaining in the beam. If a second opinion is requested, the candidate stays while higher-confidence candidates are processed first. Candidates below 0.4 are eliminated and auto-rejected with justification.

This integration ensures that Naïve Bayes handles uncertainty in a probabilistic and interpretable manner while Beam Search enforces scalability constraints by limiting the active candidate set. Together, Naïve Bayes and Beam Search maintain operational efficiency and decision transparency under resource constraints.

2.4 Conclusion

This conceptual AI hiring system combines rule-based, ontological and probabilistic knowledge to reduce ethical risks while improving decision-making under uncertainty. Rule-based filtering excludes proxy attributes; ontology neutralises discriminatory proxies and Naïve Bayes handles missing data without penalising non-linear career paths. Bayesian inference models uncertainty as probability rather than binary failure. Blinded human oversight with mandatory justifications counters automation bias. Beam search prioritises candidates efficiently under resource constraints. Together, these techniques form a coordinated pipeline that satisfies EU AI Act transparency requirements while avoiding historical bias and proxy discrimination.

(Task 2 Word Count: 1,662 words)

3.0 Appendix

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Paragraphs	17

Task 1

My Writing Details	
Words	1,662
Characters	12,370
Sentences	98
Paragraphs	34

Task 2

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